3.500 What is the probability that the top four cards of a shuffled deck are all of the same suit?

3.700 What is the probability that the top card in a shuffled deck is an Ace (of any color), and the second card is a spade? The Ace may be the Ace of spades.

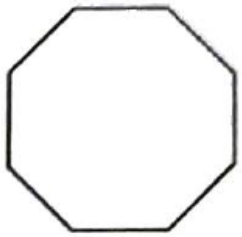
3.610 For three randomly selected positive integers, what is the probability that the sum of the units digits is even?

Example: Lisa shakes her piggy bank which contains 5 quarters and 5 nickels until 5 coins fall out. Assuming nickels and quarters fall out with equal probability, what is the probability that the sum of the change that has fallen out is \$0.85?

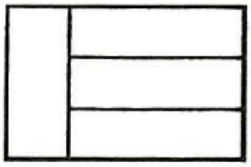
Example: There are a dozen eggs in a basket and four have been hard-boiled. You can tell by spinning an egg whether it has been hard-boiled. You select three eggs from the basket. What is the probability that only one of the three is hard-boiled?

3.030 Three numbers are selected at random and without replacement from a bag containing each of the numbers 1 through 9. What is the probability that the three numbers are consecutive integers?

3.130 Four of the vertices of a regular octagon are selected at random and connected to form a quadrilateral. What is the probability that the quadrilateral will be a rectangle?



3.230 An organization is choosing colors for the stripes on its flag. The flag has three horizontal stripes and a vertical stripe as shown. If the color of each of the four stripes is selected at random from the colors red, blue, and green. What is the probability that no two stripes of the same color share an edge?



Example: Five boys and seven girls, are randomly assigned to three groups of four students each. What is the probability that one of the three groups will be all girls?

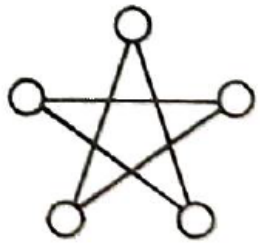
Example: The 26 letters of the alphabet are placed on tiles and randomly separated into two equal piles. What is the probability that the word MATH can be found in one of the two piles?

3.140 Three students are randomly selected from a group of twelve that includes twins Billy and Adam. What is the probability that Adam is selected but Billy is not?

3.240 A youth basketball team consists of 12 players. If they are randomly divided into 4 groups of 3 players each for passing drills, what is the probability that the 2 girls on the team will be placed in the same group?

3.450 For any date in history, Luke can quickly tell you what day of the week it was. You choose three dates at random and Luke tells you the day of the week for each. What is the probability that at least two of the dates occurred on the same day of the week?

3.750 If the digits 1 through 5 are randomly arranged so that one digit fills each of the five circles, what is the probability that at least two consecutive integers will be connected by a line segment?



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3.160 Four of the vertices of a cube are selected at random. What is the probability that all four selected points will be coplanar?

3.560 You are dealt three cards from a standard deck of 52 cards. What is the probability that exactly two cards are from the same suit?

3.660 3 digits are selected from the set {2, 3, 4, 5, 6, 7, 8} without replacement. What is the probability that the sum of the three digits is prime?

Example: The bus comes to a stop near your house every day at a random time between 6:45 and 6:50. You arrive at the bus stop at a random time between 6:40 and 6:45 every day and wait until the bus comes. What is the probability that you wait less than 4 minutes for the bus to arrive? Express your answer as a percent.

3.070 In rectangle ABCD, AB = 1 and BC = 2. Point X is selected at random within the rectangle. What is the probability that the area of triangle ABX is more than twice the area of triangle BCX?

3.370 Point X is selected at random within square ABCD of side length 3. What is the probability that quadrilateral ABXD has an area greater than 4 square units?

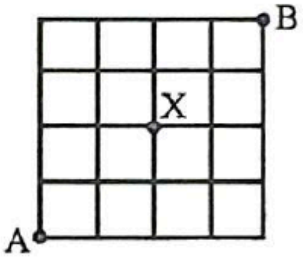
3.470 One side of a triangle is 5cm long. Two (not necessarily integer) numbers are randomly selected between 0 and 10. What is the probability that the two numbers can be the other two sides of the triangle

3.090 The expiration date on a credit card is given as two digits for the month and two digits for the year. What is the expected value for the sum of the digits in the expiration date for a credit card that expires between 01/10 and 12/15 inclusive? Assume that each month and year occur with equal frequency.

3.390 In five rolls of a standard die, what is the probability of rolling a "full house", that is, two of one number and three of a different number?

3.790 Two distinct pairs of vertices are randomly selected on a cube and each pair is connected by a line segment. What is the probability that the two line segments intersect?

3.990 Staring at point A on the map below, Jennifer needs to walk to point B using the grid of city streets shown. At each intersection, Jennifer flips a coin to decide whether to go north or east (at intersections where she has a choice). What is the probability that she will pass the intersection marked with an X on her way from A to B?



4.320 Using only the digits 1, 2, and 3 with at least one of each, what is the smallest integer that can be created which is divisible by 8 and 9?

4.520 What is the sum of all 4-digit multiples of 8 which use each of the digits 1, 2, 3, and 4 exactly once.

4.720 Each of the digits 0 through 9 is used exactly once to create a ten-digit integer. Find the greatest ten-digit number which uses each digit once and is divisible by 8, 9, 10, and 11.

4.130 The number 13 is prime, and when its digits are reversed, 31 is also prime. Besides13 and 31, how many other 2-digit primes also satisfy this condition?

4.440 If a and b represent distinct positive integers which have exactly 3 factors each, how many different products ab are less than 1,000?

4.290 Find the sum of the factors of each number below and look for a pattern to determine the sum of the factors of 2^{30}. **a.** 64 **b.** 128 **c.** 256 **d.** 2^{30}

4.390 The sum of the factors of what perfect cube is 400?

4.590 What is the least positive integer whose factor sum exceeds 100?

4.511 The GCF of two distinct perfect squares is 4, and their product is 3,600. What is the least positive difference between the two perfect squares?

4.121 Janice and Kiera begin jogging around a track, starting at the finish line and going the same direction. Janice completes a lap every 78 seconds, while Kiera takes 90 seconds to complete each lap. At the end of their workout, they cross the finish line together in a whole number of minutes for the first time. How many more laps did Janice run than Kiera?

4.321 Positive integers a, b, and c, satisfy the following conditions:

$\text{GCF}(a,b,c) = 1$.

$\text{GCF}(a,b) = 3$.

$\text{GCF}(a,c) = 4$.

$\text{GCF}(b,c) = 5$.

What is the least possible value of $\text{LCM}(a,b,c)$?

4.531 Find the largest power of 6 that divides (18!).

4.731 In the prime factorization of 2,500!, what is the power of 7?

4.341 If $n!$ ends in exactly 100 zeros, what is the greatest possible value of n?

4.451 What is the greatest number of factors that any three-digit number has?

4.061 Melody makes a list of all the proper fractions $\frac{n}{496}$ which have a numerator of 1 when written in simplest terms. What is the sum of all the positive fractions on Melody's list?

4.722 Kelsey writes a 6-digit integer. Michael copies the integer but places the units digits first, maintaining the order of the remaining five digits (for example, 123,456 becomes 612,345). If Michael's integer is fives times Kelsey's, what is Michael's integer?

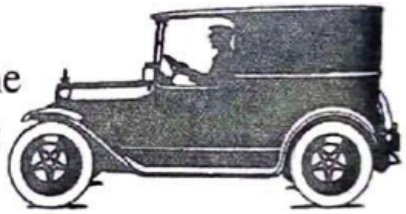
4.332 What is the remainder when $(1 + 2 + 3... + 48)^{49}$ is divided by 50?

4.142 How many of the *factors* of 900 have exactly 18 factors?

4.942 In the prime factorizations of integers a, b, and c: a and b have exactly 4 (not necessarily distinct) prime factors in common, a and c have exactly 5 prime factors in common, while b and c share 6. What is the least positive product abc for three positive integers which satisfy these conditions?

4.352 What is the greatest possible product for a set of positive integers whose sum is 25?

4.652 The tires on an antique car have wheels with five spokes, but the tires are slightly different sizes: The larger tire in back has a diameter of 30 inches, while the smaller tire in front has a diameter of 27 inches. A photographer wants to photograph the car while the wheels are in a position identical to the one above (with both stars up-side down). How many times will this occur during the course of a quarter-mile parade if the wheels begin the parade in the position shown?



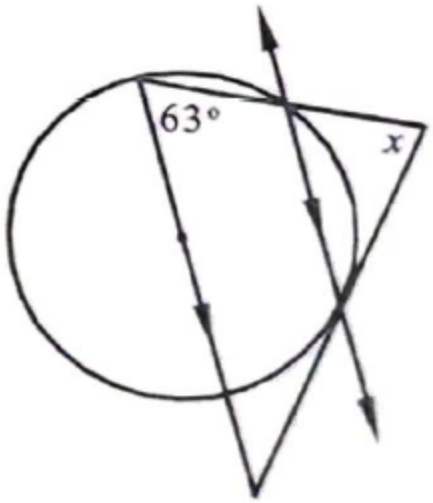
5.900 Parallelogram ABCD has side lengths 9cm and 13cm. Find the shortest integral length of diagonal AC.

5.010 What is the maximum number of diagonals in a concave nonagon (9 sides) which can be drawn entirely outside of the interior of the polygon?

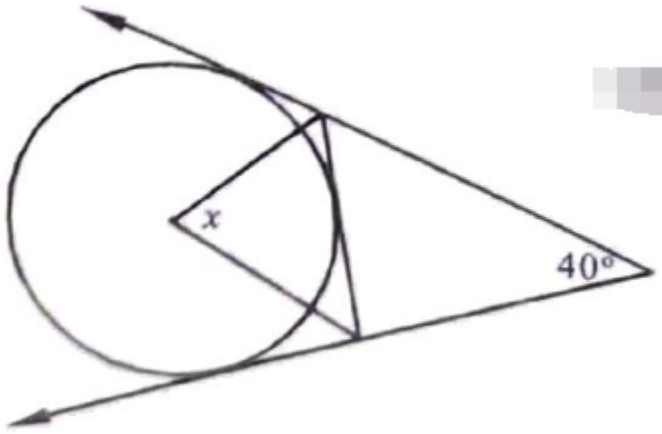
5.520 How many regular polygons have interior angles of integral measure?

5.820 The long diagonal of a rhombus is 3 times the length of the short diagonal. The perimeter of the rhombus is 40cm, what is its area? (See section 5.3).

5.730



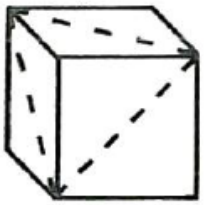
5.830



5.650 A cylinder with a 4cm diameter is 4cm tall. What is the length of the shortest path from the base to a point at the top of the cylinder on the opposite side? Express your answer in centimeters as a decimal rounded to the tenth.



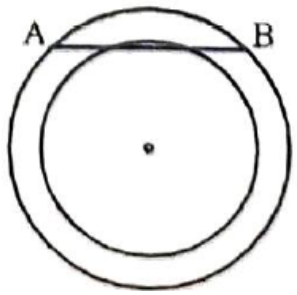
5.260 The vertices of a cube of side length 2cm are connected to form a triangle as shown. What is the area of the triangle?



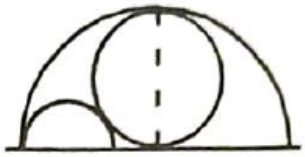
5.460 Find the length AC in regular hexagon ABCDEF if the perimeter of the hexagon is 24cm.

5.560 The midpoints of a hexagon are connected to form a smaller hexagon. What is the ratio of the perimeter of the small hexagon to the perimeter of the large hexagon?

5.270 Segment AB is trisected at its points of intersection with concentric circles of radius 7cm and 9cm. Find the length of segment AB.

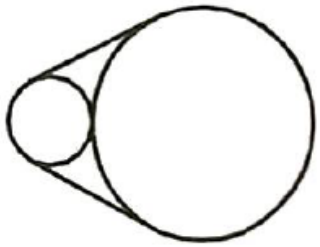


5.080 The radius of the large semicircle is 2cm. What is the radius of the small semicircle?



5.180 Square ABCD has vertices A(1,–2) and C(4, 5). What is the area of the square?

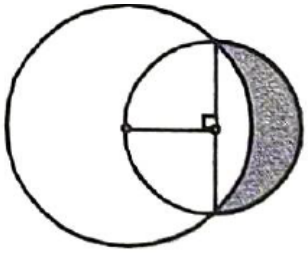
5.380 A metal band is wrapped tightly around pipes of radius 3cm and 9cm. What is the length of the band? Express your answer in simplest radical form.



5.390 A hexagon of side length 4cm has both an inscribed and a circumscribed circle. What is the area of the region (called an annulus) between the two circles?

5.490 Regular octagon ABCDEFGH has a side length of 2cm. Find the area of square ACEG.

5.790 The distance between the centers of the circles is 1cm. Find the area of the shaded region.

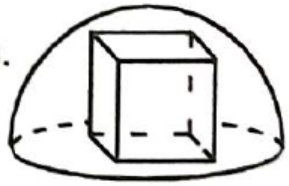


5.890 What is the smallest possible area for a triangle of integral side lengths whose perimeter is 15cm? Round your decimal answer to the hundredth.

5.990 Find the altitude to the side of a rhombus with diagonals measuring 6cm and 8cm.

5.001 A semicircle with diameter AD = 12cm is rolled to form a cone by connecting A to D. What is the volume of the cone?

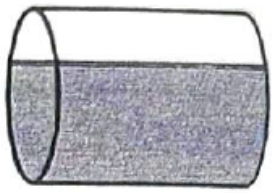
5.501 A cube is inscribed within a hemisphere. The cube has 2-inch edges. What is the surface area of the hemisphere?



5.601 A snub cube has 6 squares faces and 32 triangular faces. What is the total number of vertices and edges on a snub cube?



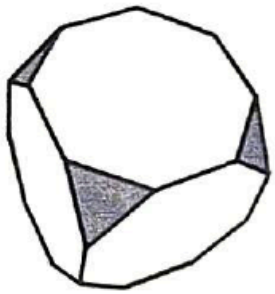
5.011 A cylinder with a 6-inch radius is laid on its side and filled to a depth of 9 inches. The cylinder is 24 inches long. What is the volume of water contained in the cylinder?



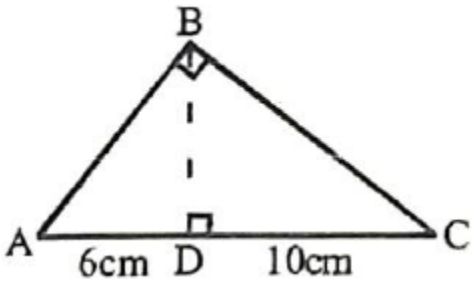
5.311 A cylinder is inscribed within a sphere. The cylinder has a 6-inch diameter and is 8 inches tall. What is the volume of the sphere?

5.411 A cube has a sphere inscribed within and a sphere circumscribed about it. What is the ratio of the surface area of the inscribed sphere to the surface area of the circumscribed sphere?

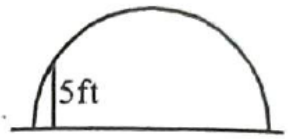
5.511 A wooden cube with 18cm edges has each of the eight corners sliced off so that the resulting faces are equilateral triangles whose edges are 6cm long. What volume has been removed from the cube? Express your answer in simplest radical form.



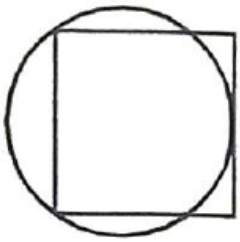
Example: The altitude to the hypotenuse of a right triangle divides the hypotenuse into segments of length 6cm and 10cm. Find the area of the right triangle.



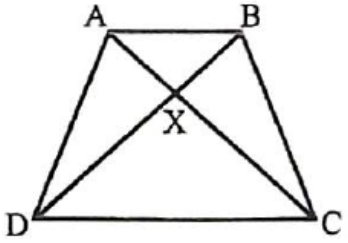
5.031 You are estimating the height of a semicircular arch. Exactly 1 foot from the base of the arch, the arch is 5 feet tall. What is the height of the arch at the center?



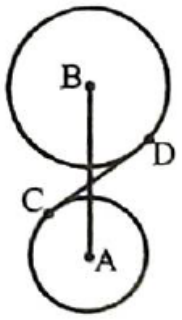
5.131 A square with 4cm sides is drawn so that its right edge is tangent to a circle, and the circle passes through two of its vertices. What is the diameter of the circle?



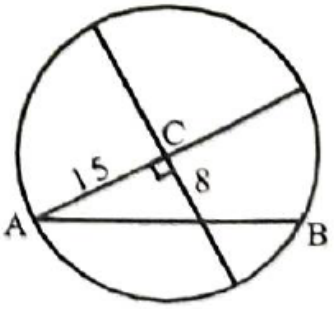
5.331 Trapezoid ABCD has base lengths AB = 30 and CD = 54cm, with congruent 56cm diagonals. Find the area of triangle ABX. Express your answer in simplest radical form.



5.431 Circle A has a radius of 28 inches, and circle B has a 44-inch radius. The distance between their centers is 90 inches. What is the length of segment CD, which is tangent to both circles?

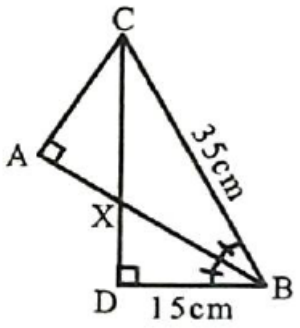


5.531 Express the length of chord AB in circle C as a common fraction in simplest form.



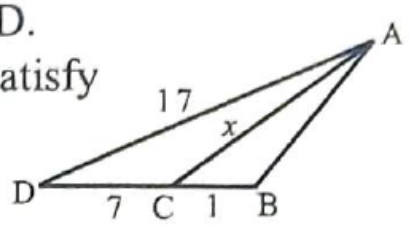
5.141 What is the length of the angle bisector to the short leg of a 9-12-15 right triangle? Express your answer in simplest radical form.

5.241 Find the length of AB in the figure. Express your answer in simplest radical form.

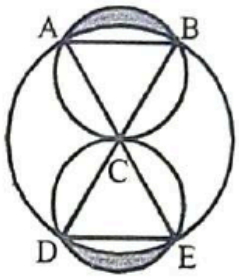


5.341 The legs of a right triangle are in the ratio 1:2. The angle bisector to the short leg divides it into two segments, one of which is 1cm longer than the other. Express the perimeter of the triangle in simplest radical form.

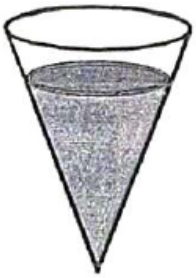
5.161 Angle B is obtuse in triangle ABD. How many integral lengths x satisfy the length of AC?



5.261 Diameters AE and BD of length 6cm form the sides of congruent equilateral triangles ABC and CDE which are circumscribed by circles. Find the combined area of the shaded lunes, which include the area inside the smaller circles but outside the larger circle.

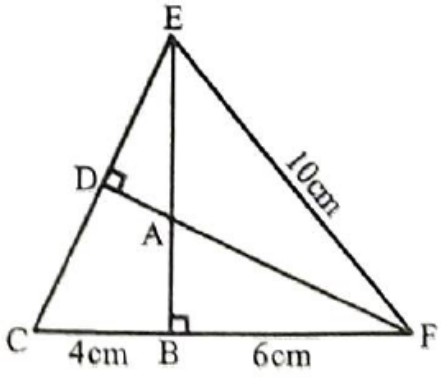


5.361 A cone has a radius of 6cm and a height of 20cm is half-full of water (by volume). What is the depth of the water in the cone to the nearest tenth of a centimeter?



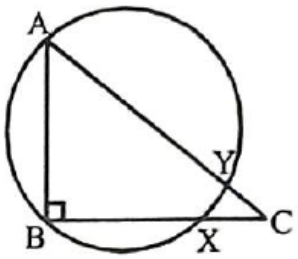
5.561 The areas of the faces of a right rectangular prism are 6, 9, and 12cm^2. To the nearest tenth of a centimeter, what is the length of the diagonal of the prism?

5.761 Two altitudes are drawn in an isosceles triangle as shown. Find the length of segment AD.

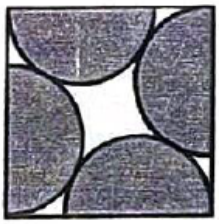


5.271 The diagonals of right trapezoid are perpendicular. What is the area of the trapezoid if the parallel bases measure 8cm and 18cm?

5.371 Right triangle ABC has points A and B on the circumference of a circle, and intersects the circle at points X and Y. Find the area of the triangle if BX = 16, AY = 32, and YC = 3.



5.471 Tangent semicircles have their diameters on the sides of a square of side length 1. Express the combined area of the four semicircles in simplest radical form.



5.571 Triangle FEC is equilateral with a side length of 20cm and FE tangent to the circle at F. Find the length of chord FA if ED = 8cm.

*Submitted by student Jason Liang.

